

Warehouse Improvement Report: An Annotated Template

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Abstract—Summarize the warehouse problem, the approaches evaluated, and the main findings. Include a decision-relevant quantitative result with its unit and comparator, then state the overall recommendation and its main limitation. Write the abstract after completing the analysis so that it reflects the final comparison. Keep it self-contained and concise.

Index Terms—warehousing, activity profiling, storage assignment, order picking

USING THIS TEMPLATE

Each section describes the content it should cover. You are free to organize the discussion within it. Replace the guidance with your analysis and update the title, authors, and affiliation. The generated evidence examples at the end demonstrate how code outputs can be included in the report.

For the abstract, IEEE recommends a self-contained paragraph of up to 250 words [1]. The UW–Madison guide explains how to connect the purpose, methods, findings, and implications [2].

I. INTRODUCTION

Introduce the warehouse and the operational questions being investigated. State the scope and the criteria used to assess proposed approaches, such as travel, space, throughput, service, and implementation effort. Explain how activity profiling informs the storage, layout, and order picking analyses, and how the final comparison weighs their benefits, limitations, and interactions. Establish the purpose of the evaluation while leaving its outcome to the analysis.

II. DATA AND EVALUATION METHODS

Describe the data, observation periods, units, and checks that support the analysis. Explain the common baseline, performance measures, assumptions, and experimental settings needed to interpret and reproduce the comparisons. Present details specific to an approach where that approach is evaluated. Cite the sources supporting the methods and make the reported results traceable to the saved evidence [3]. Keep installation and execution instructions in the README.

III. WAREHOUSE ACTIVITY PROFILING

Characterize the current operation using warehouse activity, customer-order, item-activity, and inventory profiles, together with relevant performance indicators. Explain the patterns, imbalances, or constraints revealed by the evidence and their implications for subsequent design choices. Define the scope,

units, and calculation of each measure. Select tables and figures that support the diagnosis and explain what they show.

IV. STORAGE ASSIGNMENT AND SPACE REQUIREMENTS

Explain the proposed storage approaches and how they determine product locations and space requirements. Describe the model, parameters, assumptions, and constraints needed to evaluate them. Compare their performance with the current assignment on a consistent basis, supported by quantitative results and a readable layout. Discuss capacity, compatibility, replenishment, and service, and identify the conditions under which each approach is suitable.

V. WAREHOUSE LAYOUT DESIGN

Explain the layout alternatives and the space, storage, throughput, and handling requirements they address. Show the arrangements and flows clearly, with the scale and assumptions needed to interpret them. Evaluate relevant performance measures against the baseline and discuss operational feasibility, including access, safety, replenishment, staging, and equipment. Explain the benefits, limitations, and implementation implications of the alternatives.

VI. ORDER PICKING SYSTEM DESIGN

Describe the proposed picking approaches and why they suit the observed workload. Explain the relevant process rules, resources, and interactions with storage and layout. Evaluate the alternatives under comparable operating conditions using appropriate performance measures. Discuss uncertainty, service, replenishment, and implementation effort, and explain the conditions under which the approaches are feasible and useful.

VII. PERFORMANCE EVALUATION AND TRADE-OFFS

Bring together the storage, layout, and picking findings using consistent baselines, demand, units, and constraints. Weigh their effects on performance, space, service, workload, cost, and risk where the evidence permits. Examine how the approaches interact when combined, including benefits or burdens shifted between processes. Explain how uncertainty and data limitations affect the comparison, then justify a suitable combination or sequence of changes and the conditions for adopting it.

Table I illustrates a comparison with the baseline. Choose measures relevant to your analysis, define their scope and

units, and replace the placeholders with values generated by your code. Adapt the rows to the alternatives evaluated. Define the service measure and cost components in the text.

TABLE I
A COMMON SET OF MEASURES SUPPORTS COMPARISON OF THE BASELINE AND PROPOSED INTERVENTIONS.

Option	Travel (m/order)	Service (%)	Cost (EUR/order)
Baseline	[value]	[value]	[value]
Storage	[value]	[value]	[value]
Layout	[value]	[value]	[value]
Picking	[value]	[value]	[value]
Combined	[value]	[value]	[value]

VIII. CONCLUSION

Answer the overall evaluation question using the strongest findings. State the recommended course of action, the main quantitative evidence supporting it, and the conditions or limitations that matter to the decision. Identify any further evidence or monitoring needed before implementation. Keep the conclusion consistent with the preceding analysis.

GENERATED EVIDENCE EXAMPLES

The following tables and chart demonstrate the connection between the code and the report. Replace them with evidence relevant to your analysis. The S. P. Richards data describe a Philadelphia distribution center [4]; the inventory quantities in this example are dated 1 February 2005. Table II records the example settings. Use units in table headings and chart axes, and captions that state the finding and its scope.

TABLE II
THE INVENTORY BASELINE AND ILLUSTRATIVE SCENARIO USE THE MEASURES AND ASSUMPTIONS LISTED HERE.

Setting	Value
<i>Inventory data</i>	
Source	inventory in active locations
Records	20,804
Zones	5
<i>Measures</i>	
Pick locations	active picking locations by zone
SKUs	distinct SKUs by zone
Inventory	selling units on hand
<i>Illustrative scenario</i>	
Inventory multiplier	1.10
Randomness	none

The inventory snapshot covers 20,804 active picking locations across 5 zones and 1,172,671 selling units on hand (Table III). On-hand inventory does not indicate pick frequency.

TABLE III
IN THE PHILADELPHIA DC, ZONE C HAS THE MOST ACTIVE PICKING LOCATIONS, WHILE ZONE B HOLDS THE MOST SELLING UNITS ON HAND.

Zone	Active picking locations	Distinct SKUs	Selling units on hand
A	4,996	4,996	181,427
B	5,614	5,614	682,057
C	8,961	8,961	279,135
F	697	697	14,879
L	536	536	15,173
All zones	20,804	20,804	1,172,671

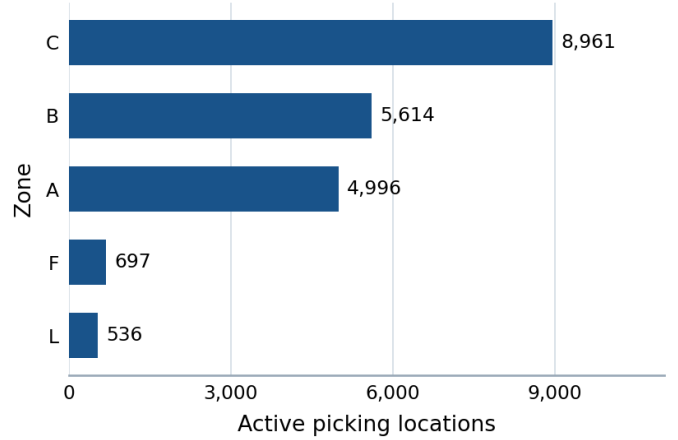


Fig. 1. Zone C contains 43.1% of the Philadelphia DC's 20,804 active picking locations.

Table III and Fig. 1 show how inventory is distributed across zones. Active picking locations and selling units on hand rank the zones differently. Demand, travel time, and available capacity need separate data to evaluate storage policies.

Table IV compares observed on-hand inventory with a quantity multiplied by the same factor in each zone. The calculation changes recorded quantities without changing warehouse operations. It demonstrates how to present a generated comparison. A policy experiment should change a defined operating decision and evaluate its consequences.

TABLE IV
THE ILLUSTRATIVE TOTAL EXCEEDS THE OBSERVED TOTAL BY 10.0% AFTER MULTIPLYING EACH ZONE'S SELLING UNITS ON HAND BY 1.10 AND ROUNDING UP.

Zone	Selling units on hand		
	Observed	Illustrative	Difference
A	181,427	199,570	18,143
B	682,057	750,263	68,206
C	279,135	307,049	27,914
F	14,879	16,367	1,488
L	15,173	16,691	1,518
All zones	1,172,671	1,289,940	117,269

The example holds active picking locations and SKUs fixed. Its difference in selling units on hand does not estimate changes in picking effort, storage capacity, or service.

REFERENCES

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- [3] The Turing Way Community, “Research compendia,” The Turing Way. [Online]. Available: <https://book.the-turing-way.org/reproducible-research/compendia/>
- [4] J. J. Bartholdi, III and S. T. Hackman, “A Project for S. P. Richards Company,” Warehouse and Distribution Science. [Online]. Available: <https://www.warehouse-science.com/wh/projects/spr3/main.html>